KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Appointment Scheduling and Reminder System

Submitted by

S SARVESH
25BCS146

Under the Guidance of

Mr. ARAVIND

Technical Trainer

Technical Department, KG College of Arts and Science

Academic Year:**2025 – 2026**

APRIL 2026

Semester: II

DECLARATION

I hereby declare that the project entitled "**Appointment Scheduling and Reminder System"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of **Mr.ARAVIND**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

Ihave acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

S SARVESH

CERTIFICATE

This is to certify that the project entitled "**Appointment Scheduling and Reminder System**" is a bona fi de work carried out by **SARVESH S I "B"/2025, B SC COMPUTER SCIENCE,** KG College of Arts & Science, in partial fulfi llment of the requirements for the course **JAVA PROGRAMMING.**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:10.04.2026

Submitted for the Viva-Voce Examination held on 10/04/2026

Guide:
(Technical Head, KGCAS)

Mr.Aravind

Head of the Department (Trainer, KGCAS)

Mr.Booplan Sundaramoorthy

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	

3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	

6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

Abstract:

The Appointment Scheduling and Reminder System is designed to automate the process of booking, managing, and scheduling methods often lead to missed appointments, double bookings, and inefficient time management.

This system provides a digital solution where users can book appointments easily through an interface, and the system errors, and improves customer satisfaction.

The system maintains records of all appointments, helps in tracking schedules, and provides reports for better decision-making. Overall, it enhances efficiency, accuracy, and reliability in appointment management.

1.INTRODUCTION

schedules. To overcome these issues, an Appointment Scheduling and Reminder System is developed to automate the

Managing appointments manually is time-consuming and prone to errors such as overlapping bookings and missed
entire process.
This ensures smooth scheduling and better time management.

This system allows users to book, update, or cancel appointments easily while sending timely reminders to avoid missing them.

It improves organization, saves time, and enhances productivity.

To automate the appointment booking process.

To reduce scheduling conflicts and human errors.

To send automatic reminders to users.

To maintain a centralized database of appointments.

To improve efficiency and user satisfaction.

1.1 Objectives

1.2 Overview

The system integrates three major components:

1.Appointment Booking Module – Users can select date and time slots and book appointments.

2.Reminder Module – Sends notifications via SMS/email/app alerts before the appointment.

3.Database & Reporting – Stores appointment data and generates reports.

1.3 Key Features

Online AppointmentBooking– Users can book appointments easily by selecting date and time.

Real-Time Slot Availability– Supports booking via app, website, or system.

Automated Reminders– Sends alerts before appointments to reduce misses.

Multiple Booking Options– Shows available slots and avoids double booking**.**

User Management – Stores user details and appointment history.

Schedule Management– Helps manage, update, or cancel appointments.

Reports & Analytics– Provides basic insights on appointments and usage.

User-Friendly Interface– Easy for users and admins to operate.

Security & Data Backup– Keeps data safe with backup support.

2.PROGRAM ANALYSIS

Programanalysisisanimportant phase in software development that focuses on understanding the system requirementsandidentifying the problems in the existing system. It involves a detailed study of how the current systemoperates,whatlimitations it has, and what improvements are needed. This phase helps in gathering user requirements,analyzingworkflow, and defining the functionalities that the new system should provide.

2.1 Problem Definition

Manual appointment systems often lead to several issues that affect efficiency and time management. One of the major problems is missed appointments, which occur due to the lack of proper reminder mechanisms. In addition, there is a high possibility of double booking of time slots, especially when multiple appointments are handled manually without proper tracking. Managing schedules also becomes difficult, as there is no centralized system to organize and monitor appointments effectively. This results in confusion, delays, and poor coordination between users and service providers. Furthermore, manual handling leads to poor time management, reducing overall productivity and customer satisfaction.

2.2 Project Overview

The system securely stores all appointment details, including user information, date, time, and service type, in a centralized database. It efficiently manages schedules by preventing overlapping bookings and ensuring proper time allocation. In addition, administrators can easily monitor, update, and control appointments through a dedicated interface, making the management process more organized and efficient. Overall, this system improves accuracy, saves time, enhances productivity, and provides a better user experience for both customers and administrators.

2.3 Scopes and Limitations

Scope

Applicable for hospitals, salons, offices, etc.
Supports online booking and cancellation
Provides reminders and reports

Limitations

Requires internet connectivity

Limited customization for complex needs

Data security depends on implementation

3.SYSTEM DESIGN

System design is an important phase in software development that defines how the system will be structured and implemented. It involves planning the overall architecture, components, modules, and their interactions performing aspecific function, which helps in better organization and easier development.

3.1 Modules

The main modules of the **Appointment Scheduling and Reminder System** are explained below:

1. Appointment Booking Module

1.Allows users to select date and time

2.Displays available slots

3.Confirms booking instantly

2. Reminder Module

1.Sends notifications before appointment

2.Supports SMS/email alerts

3.Reduces missed appointments

3. Database Module

1.Stores user and appointment details

2.Maintains history of bookings

3.Ensures data security

4. User Management Module

1.Handles user login and registration

2.Manages user profiles

5. Exit Module

1.Safely exits the system

2.Saves all data

4.METHODOLOGY

Methodology explains the step-by-step process followed to develop the system. It includes all stages from understanding the problem to designing, developing, testing, and implementing the system. A proper methodology ensures that the project is completed in an organized and efficient manner while meeting user requirements.

4.1 Problem Analysis

1.Identified issues in manual scheduling such as missed appointments, double booking, and lack of reminders.

2.Studied user requirements by understanding how users interact with the system and what features they expect.

3.Analyzed system needs to determine essential functionalities like booking, reminders, and data management.

4.Examined limitations of the existing system and proposed suitable solutions.

5.Defined clear objectives to improve efficiency and reduce errors.

4.2 System Design

1.Designed modules for appointment booking, reminder system, and database management.

2.Defined the workflow of the system, including how data flows between different modules.

3.Created system architecture to ensure proper interaction between components.

4.Planned user interface design for easy and smooth user experience.

5.Structured the system in a modular way to improve scalability and maintenance.

4.3 Program Development

1.Developed the system using programming languages such as Java, Python, or C#.

2.Created a user-friendly interface using GUI or web technologies.

3.Integrated all modules to work together as a complete system.

4.Implemented database connectivity to store and retrieve appointment data.

5.Ensured the code is efficient, maintainable, and error-free.

4.4. Testing and Validation

The methodology explains the steps followed to develop the system in an organized way. It includes problem analysis, system design, and program development. This process ensures the system works efficiently and meets user requirements.

Example Test Cases:

Test Case	Action	Result
1. Booking appointment	Schedule an appointment	Success
2. Sending reminder	Send appointment reminder	Delivered
3. Cancel appointment	Cancel the appointment	Updated
4. Other Test Case	Other action steps...	N/A
5. Other Test Case	Other action steps...	N/A

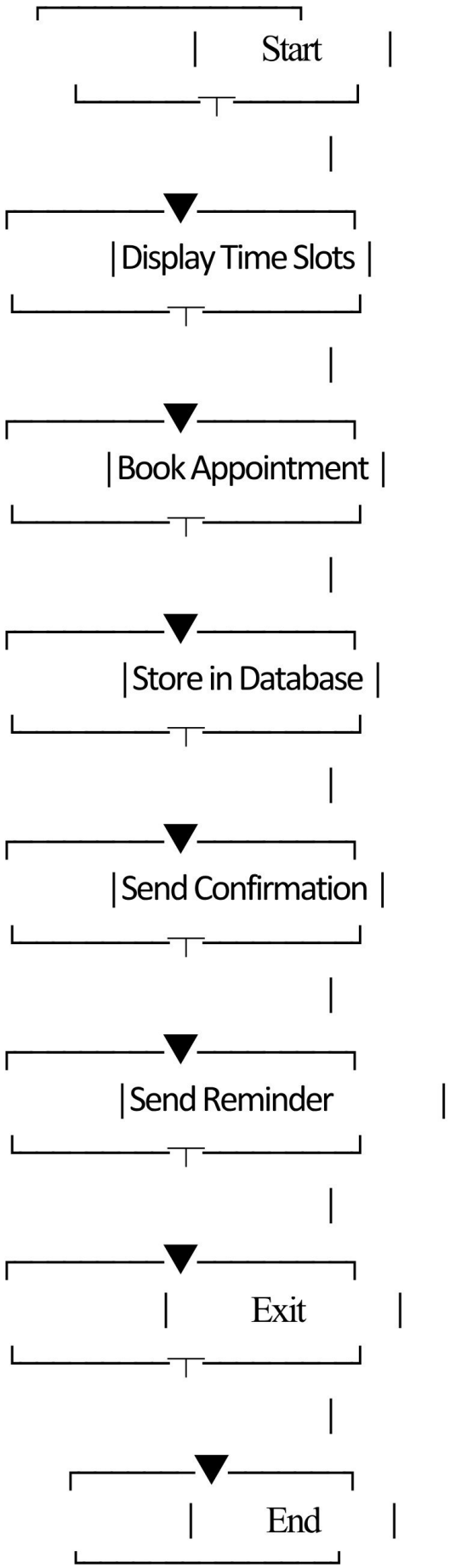
4.5. Result and Documentation

The result and documentation phase shows that the Appointment Scheduling and Reminder System works efficiently by automating the process of booking and managing appointments. It helps in reducing errors such as double booking and missed appointments, while also minimizing delays in scheduling. The system ensures smooth operation and improves overall performance. Additionally, proper documentation is maintained, including system design, features, and usage details, which makes it easy to understand, use, and maintain the system in the future.

4.6. Algorithm

Algorithm for the **Appointment Scheduling and Reminder System**

4.7. Flowchart



Source code

```
import java.util.*;


public class AppointmentSystem {

static Scanner sc = new Scanner(System.in);


    public static void main(String[] args) {

        System.out.println("Appointment Scheduling System");


        System.out.print("Enter your name: ");

        String name = sc.nextLine();


        System.out.print("Enter appointment date: ");

        String date = sc.nextLine();


        System.out.print("Enter time: ");

        String time = sc.nextLine();

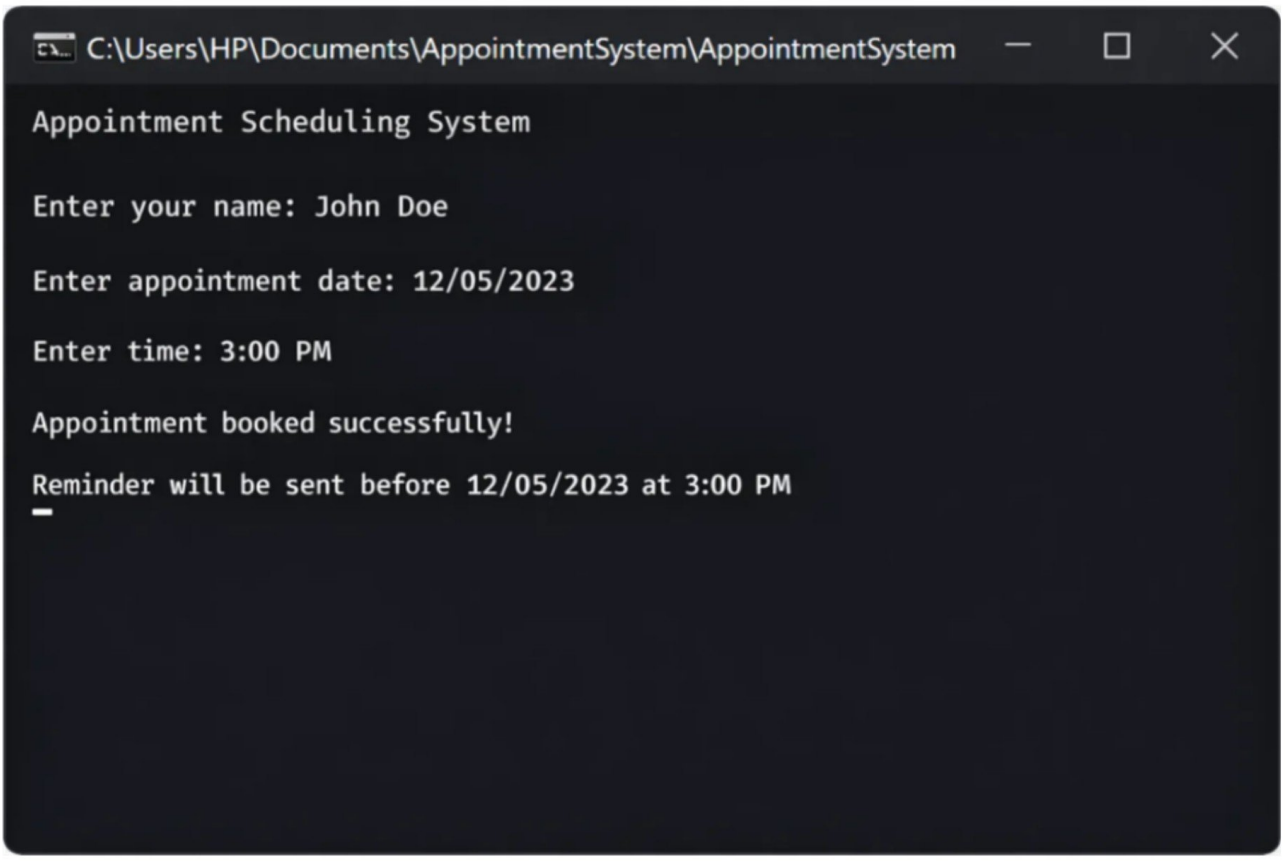

        System.out.println("Appointment booked successfully!");

        System.out.println("Reminder will be sent before " + date + " at " + time);

    }

}
```

6.INPUT AND OUTPUT SCREENS



7.DISCUSSION & CONCLUSION

7.1 Discussion

The Appointment Scheduling and Reminder System successfully reduces manual effort and improves scheduling efficiency by automating the entire booking process. It ensures proper time management by organizing appointments systematically and sending timely reminders, which helps in reducing missed appointments. The system also improves coordination between users and administrators, making the overall process smooth and reliable.

In addition, the system enhances productivity by minimizing human errors such as double booking and incorrect scheduling. The use of a centralized database allows easy tracking and management of appointment records. Users find the system convenient as it provides quick access to booking and updates. However, the system depends on stable internet connectivity for smooth operation and requires proper setup and user training. Despite these limitations, the system proves to be highly useful in real-time applications.

7.2 Conclusion

The Appointment Scheduling and Reminder System is an efficient and reliable solution for managing appointments in a digital way. It improves accuracy by reducing manual errors and ensures better time management through proper scheduling and reminders. The system enhances user experience by providing a simple and user-friendly interface for booking and managing appointments.

Furthermore, the system is scalable and flexible, making it suitable for various industries such as healthcare, education, and business organizations. It helps organizations improve productivity, save time, and provide better service to users. Overall, the system plays an important role in modernizing appointment management and contributes to increased efficiency and customer satisfaction.

8.REFERENCES

The following resources were used during the development of this project:

Books

1.Pressman, Roger S. Software Engineering: A Practitioner's Approach. McGraw-Hill Education, 8th Edition, 2014.

2.Sommerville, Ian. Software Engineering. Pearson Education, 10th Edition, 2015.

Websites

W3Schools. "Java Programming Tutorial."

Available at: https://www.w3schools.com/java

Oracle Corporation. "Java Platform, Standard Edition Documentation."

Available at: https://docs.oracle.com/javase